

SEMI SUPERVISED FRAMEWORK FOR ERROR DEBUGGING IN DIGITAL VLSI DESIGN USING DEEP LEARNING

*Report submitted to the SASTRA Deemed to be University
in partial fulfillment of the requirements
for the award of the degree of*

B-Tech Electronics & Communication Engineering

Submitted by

DHARSHINI S

(Reg. No: 226004016, B-Tech Electronics & Communication Engineering)

RASHMI K N

(Reg. No: 226004083, B-Tech Electronics & Communication Engineering)

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Bonafide Certificate

This is to certify that the project report titled “**Semi Supervised Framework for Error Debugging in Digital VLSI Design Using Deep Learning**” submitted in partial fulfillment of the requirements for the award of the degree of B. Tech. Electronics and Communication Engineering to the SASTRA Deemed to be University, is a bona-fide record of the work done by **Ms. Dharshini(Reg. No. 226004016), Ms. Rashmi(Reg. No. 226004083)** during the final semester of the academic year 2025- 26, in the **Department of Electronics and Communication Engineering, Srinivasa Ramanujan Centre, SASTRA Deemed to be University, Kumbakonam** under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

Signature of Project Supervisor:

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Examiner 1

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Declaration

We declare that the project report titled “**Semi Supervised Framework for Error Debugging in Digital VLSI Design Using Deep Learning**” submitted by us is an original work done by us under the guidance of **Dr. Srinivasan A, Associate Professor, Department of Electronics and Communication Engineering, Srinivasa Ramanujan Centre, SASTRA Deemed to be University, Kumbakonam** during the final semester of the academic year 2025-26, in the **Srinivasa Ramanujan Centre SASTRA Deemed to be University, Kumbakonam**. The work is original and wherever We have used materials from other sources, we have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

Signature of the candidate(s) :

- 1.
- 2.

Name of the candidate(s) :

1. Dharshini S
2. Rashmi K N

Date :

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ABSTRACT

This project presents the design and implementation of a semi-supervised framework for error debugging in digital VLSI circuits using deep learning techniques. The proposed system replaces traditional SAT-based debugging approaches with a data-driven model that reduces computational complexity and improves scalability. A Variational Autoencoder (VAE) is employed for unsupervised feature extraction, transforming raw ATPG test patterns into meaningful latent representations. These features are then processed by a hybrid classifier that adapts its architecture using Convolutional Neural Networks (CNN) for combinational circuits and Bidirectional LSTM (Bi-LSTM) for sequential circuits to accurately detect and localize faults. To further enhance performance, class imbalance handling and per-fault threshold tuning are incorporated, ensuring reliable multi-label fault prediction. The design achieves high accuracy, faster debugging speed, improved fault coverage, and efficient resource utilization. The proposed approach provides an effective and scalable solution for modern high-performance VLSI design and testing applications.

Keywords: *Deep Learning, VLSI Fault Debugging, ATPG Test Patterns, Multi-label Classification, Fault Localization.*

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ABBREVIATIONS

1D-CNN	: 1-Dimensional Convolutional Neural Network
AI	: Artificial Intelligence
ATPG	: Automatic Test Pattern Generator
AUC	: Area Under Curve
Bi-LSTM	: Bidirectional Long Short-Term Memory
CNN	: Convolutional Neural Network
CPU	: Central Processing Unit
GPU	: Graphics Processing Unit
IC	: Integrated Circuit
ISCAS	: International Symposium on Circuits and Systems (Standard Benchmark)
KL	: Kullback-Leibler (as in KL Divergence)
MCC	: Matthews Correlation Coefficient
MUS	: Minimal Unsatisfiable Subset
ROC	: Receiver Operating Characteristic
SAT	: Boolean Satisfiability
SoC	: System-on-Chip
VAE	: Variational Autoencoder
VLSI	: Very Large-Scale Integration

CHAPTER 1

INTRODUCTION

1.1 Background of VLSI Design and Verification

There have been many improvements in IC technology over the years and the reason for these improvements has been the development of semiconductor technology. Presently, with the advancements in Very Large-Scale Integration (VLSI), billions of transistors can be packed on a single chip allowing better performance of the devices.

This increase in complexity not only complicates the design process but also increases the likelihood of errors in the IC design flow. During pre-silicon verification (PSV), it is critical to make sure that there will be no fault in the designed hardware before its manufacture. Missing out even one mistake at this stage could result in expensive modifications during the post-manufacturing phase of IC development. Thus, the need for thorough testing and fault detection becomes apparent at this juncture.

1.2 Limitations of Traditional Fault Detection

In previous techniques, fault detection and diagnosis was based on mathematical approaches, which involved determining the causes of the faults in digital circuits by representing their behaviours in the form of Boolean Satisfiability (SAT). By applying SAT solvers, one can determine the existence of the faults in the circuit, such as the presence of a “stuck-at” fault where the signal line can only be 0 or 1.

Despite its accuracy and reliability, the SAT technique suffers from an inherent drawback associated with complexity. When dealing with increasingly complex circuits, the number of possibilities will increase exponentially, resulting in the phenomenon known as search space explosion, where computing is significantly heavier and requires much time. Hence, using SAT solvers repeatedly for fault detection and isolation

1.3 The Shift Towards Data-Driven and Deep Learning Approaches

To address the limitations of traditional mathematical methods, the industry and research community are increasingly moving towards data-driven approaches for fault detection. Instead of relying only on complex calculations, machine learning and deep learning techniques treat fault detection as a pattern recognition problem.

In this approach, models learn from large amounts of past test data, including both faulty and fault-free cases. Rather than analyzing every possible logic combination, these models identify patterns and relationships in the data to detect faults more efficiently. Deep learning methods, especially autoencoders, are particularly effective because they can handle high-dimensional test data and reduce it into a smaller set of important features. This helps in reducing complexity, speeding up the fault detection process, and maintaining high accuracy at the same time.

1.4 Problem Statement

While there is huge potential in the use of artificial intelligence techniques to debug hardware, most data-based systems still find it difficult to find a compromise between speed and precision. This problem becomes more acute when working with large sequential designs with tens of thousands of input/output pins.

The primary focus of the proposed system is on building an automated approach to debugging errors during hardware design which is able to work without the heavy cost associated with SAT solvers while maintaining high fault detection precision. Specifically, the key challenge is related to developing a completely unsupervised approach to extracting meaningful features from the input data without overfitting on them. In addition, the proposed system will require a highly efficient multi-label classifier capable of recognizing several stuck-at faults simultaneously.

1.5 Motivation

The primary motivation of this project is to address the key issues that arise in pre-silicon diagnosis caused by the increasing complexity of today's VLSI chips. As the size and complexity of the circuit increase, the classical approach based on SAT solvers for fault

localization struggles with the issue of search space explosion, which slows down the overall computation.

This research aims at mitigating the issue of delay by circumventing an exhaustive mathematical search while ensuring highly accurate fault detection. This will be achieved through the proposed novel hybrid deep learning methodology. The resulting benefit will be the simplification of the VLSI design process, allowing faster design-to-tape-out flow.

1.6 Hybrid Framework for Fault Diagnosis

As described above, one of the main issues associated with SAT-based debugging is the extremely high computational cost. To address this problem, a two-staged hybrid architecture is proposed which utilizes the strengths of various types of deep neural models for VLSI fault diagnosis purposes.

- **Latent Features Extraction via VAE:** Rather than using raw test data for classification purposes, our system employs Variational Autoencoders (VAEs). Using this model, the input data is compressed in the form of latent variables representing all the necessary information about circuit faults without additional noise.
- **Classification of Faults in Combinatorial Circuits via 1D-CNN Model:** For classification of combinational circuits, 1D-Convolutional Neural Networks (1D-CNNs) will be used to extract relevant patterns. Such a network will allow detecting correlations between logical elements.
- **Classification of Faults in Sequential Circuits via Bi-LSTM Networks:** For classification of sequential circuits, Bi-LSTM neural networks will be used. Unlike combinational circuits, they include state elements that require temporal processing. Combining the results obtained at the first stage of the process with the output data from the second stage of classification, we obtain a multi-label classifier system

CHAPTER 2

OBJECTIVES

The primary objective of this project is to build an efficient Hybrid Deep Learning algorithm that will be able to detect and locate the presence of stuck-at faults within the VLSI circuits. The focus here is to shift from conventional mathematically-based approaches to adopt a machine learning solution that will increase the speed of locating faults within the circuits without compromising the accuracy rate.

To achieve these objectives, the project has been broken down into the following objectives:

- **For avoiding the “Search Space Explosion” problem:** Design a fault diagnosis solution that does not rely heavily on SAT solvers. In this way, the time consumed in identifying faults within the combinational and sequential circuits will be greatly reduced.
- **For feature extraction using a Variational Autoencoder:** Apply Masked Latent Regularization within a Variational Autoencoder in order to map test data onto lower dimensional features.
- **To design dual path classifier:** Construct a hybrid system, which includes 1-Dimensional Convolutional Neural Network (1D-CNN) for detecting spatial patterns in combinatorial circuits and Bi-directional LSTM (Bi-LSTM) network for understanding the temporal dynamics of sequential circuits.
- **For high accuracy and fault coverage:** Evaluate the developed system using a benchmark test circuit, to prove the robustness of the designed system. This system will have a very high accuracy and other performance measures even for simultaneous presence of multiple faults.

In summary, this research proposes an efficient and effective fault diagnosis system for better VLSI design and debugging.

CHAPTER 3

LITERATURE SURVEY

Table 3.1: Summary of Literature Survey

YEAR	AUTHOR	TITLE	KEY FINDINGS
2022	Moness et al.	Automated Design Error Debugging of Digital VLSI Circuits	Deep Sparse AE + Classifier
2021	Gaber et al.	Fault Detection based on Deep Learning for VLSI Circuits	Semi-supervised model; SSAE feature extraction
2020	Gaber et al.	Computation of Minimal Unsatisfiable Sub formulas for SAT-Based Error Diagnosis	GPU-accelerated MUS extraction; SAT-based baseline
2019	Gaber et al.	Improved Automatic Correction for Digital VLSI Circuits	MCS-based auto-correction; formal verification baseline
2018	El Mandouh & Wassal	Application of ML in Post-Silicon Debugging and Bug Localization	ML for post-silicon bug localization; need for pre-silicon methods
2018	Shi et al.	Autoencoder + LSTM Hybrid for Rare Fault Event Detection	AE for anomaly detection + LSTM for fault type classification
2017	Rodriguez Gomez	Machine Learning Support for Logic Diagnosis	SVM-based fault localization; traditional ML limits
2009	Vincent et al.	Extracting and Composing Robust Features with Denoising Autoencoders	Denoising AE theory; foundation for sparse AE variants

CHAPTER 4

TOOLS AND ENVIRONMENT

For the development, training, and evaluation of the proposed hybrid deep learning approach, the appropriate software tools and computing environment must be provided. This paper utilizes a state-of-the-art cloud-based system in conjunction with machine learning, data processing, and circuit simulation libraries.

4.1 Required Software and Programming Languages

- **Google Colab (Notebooks):** Google Colab was used for all the stages of designing a deep learning pipeline including data preprocessing, learning, and performance testing. It was chosen due to its user-friendly interface, simplicity of operation without installing any software, and possibility to conduct research in the familiar form of interactive Jupyter notebooks. Moreover, there is no need to set up hardware and it allows using cloud resources that are quite efficient for training deep learning algorithms.
- **ATALANTA ATPG Tool:** For preparing circuit data and generating necessary inputs for the deep learning algorithm, the ATPG tool called ATALANTA was employed. As described in Chapter 2, the tool processes digital circuits, simulates their behavior and generates test patterns. Moreover, together with the test vectors, it computes the corresponding stuck-at faults which become the input data to learn from.

4.2 Libraries and Frameworks

- **TensorFlow and Keras:** This framework serves as the base of the deep learning model used in this project. The Keras library, which is compatible with TensorFlow and supports execution in Google Colab, was used to design the whole model structure. Specifically, Keras is utilized in building the Variational Autoencoder (VAE) for feature extraction, and Bi-LSTM and Conv1D layers for classification. Additionally, these frameworks manage essential operations like model training, optimization, loss function definition, and backpropagation.
- **NumPy and Pandas:** These tools were employed in pre-processing and handling the dataset. NumPy was applied to transform the raw patterns of the tests provided by the ATALANTA tool to numerical vectors that could be fed to the neural network. Pandas

was applied to arrange the information in tabular format (data frame) for better storage and analysis of the results for various circuit benchmarks.

- **Scikit-learn (sklearn):** This library was used to assess the model's performance. Specifically, this framework offers a number of built-in functions to measure key metrics like accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), Hamming Loss, and ROC-AUC.
- **Matplotlib and Seaborn libraries:** These two libraries are useful in creating various graphs including confusion matrices, loss curves, performance heat maps, and radar diagrams. Through their use, one can easily analyze the performance of the algorithm as well as its ability to detect faults.

4.3 Hardware Environment (Cloud Acceleration)

Given the considerable demand for computational power involved in deep learning model training, specifically the optimization process of the VAE combined with the Bi-LSTM and CNN-based classifiers, using the normal computer CPU cannot achieve desirable results.

- **GPU Processing through Cloud Services:** This project leverages the cloud-based Graphics Processing Units (GPUs) provided by Google Colab to accelerate the training process. Through parallel execution of operations, the GPUs reduce the duration of each training epoch. Therefore, the entire training process becomes much more effective.
- **Memory Management:** In order to avoid memory issues arising during the training of big circuit datasets, the TensorFlow dynamic memory growth function was enabled within the codes. Such function will effectively manage GPU memory allocation and avoid memory overflow issues that may arise when processing large sequential circuits such as ISCAS'89.

CHAPTER 5

METHODOLOGY

The methodology chapter describes the general approach for the development of the proposed automated debugging system. Instead of applying the traditional SAT-solving approach, this project implements a data-driven methodology. It involves three major steps: collecting and processing the data, extracting features unsupervised, and implementing hybrid classification.

5.1 Data Collection and Preprocessing

Deep learning algorithms require high-quality training data as the foundation of their performance. In this study, the training dataset comprises artificially generated test vectors together with their respective masks of faults. Given that there is no need for hardware testing, the ATALANTA Automatic Test Pattern Generation (ATPG) tool is utilized for data generation. It requires standard benchmark circuit files as input and outputs the test patterns, which detect either stuck-at-0 or stuck-at-1 faults.

After the .pat files have been obtained, a specialized Python script parses the Boolean textual data into numerical arrays. This step is crucial as neural networks cannot handle unprocessed textual descriptions of circuits.

One important aspect of this step involves sorting the data depending on the kind of circuit that it is:

- **Combinational Logic:** Due to the fact that combinational logic circuits rely on current inputs only, the data is organized using 1-D arrays only.
- **Sequential Logic:** In sequential logic circuits, on the other hand, the data has to be sorted using sequences, given that such circuits rely on their past states and clock cycles.

5.2 Unsupervised Feature Extraction (Variational Autoencoder)

Using a large number of patterns directly in the form of raw data for training a classifier is likely to cause overfitting in particular in cases involving a highly complicated VLSI circuit

structure. The first component of the proposed solution utilizes a Variational Autoencoder (VAE) to address this problem.

While a regular autoencoder learns some fixed compression of data, the variational autoencoder describes the probability distribution with its mean and standard deviation. It allows for a better learning of the underlying patterns within the data.

In training the VAE model, the encoder compresses the input data into a smaller latent space. Then, the samples are obtained via the reparameterization technique from the probability distribution of the learned representation and fed to the decoder which attempts to reconstruct the initial input.

Training is carried out by minimizing a couple of losses. First, the reconstruction error is calculated as the difference between the original data point and the reconstructed one. The second loss corresponds to Kullback-Leibler divergence and aims at smoothing out the learned distribution.

In effect, the VAE model will be able to extract only critical fault characteristics and eliminate irrelevant and redundant information.

5.3 Hybrid Classification Architecture (CNN & Bi-LSTM)

The last step of the methodology involves the process of classification, where the precise fault locations in the circuit are detected. Given that combinational and sequential circuits operate differently, a dual path hybrid architecture is required for their classification. The proposed model takes two inputs simultaneously; namely, the test sequences and the compressed features extracted from the VAE.

- **For Combinational Circuits:** The raw input features are filtered through 1D-CNN layers, which traverse the entire dataset to detect local features and signal dependencies among nearby elements. It is particularly relevant in combinational logic circuits because it captures the effect of the interaction between input pins.
- **For Sequential Circuits:** The sequential input is processed via the Bi-LSTM model, which scans input data both backwards and forwards to capture the impact of previous states on the future output values. It plays an essential role in identifying faults in memory units, such as flip-flops.

After that, the outputs from either CNN or Bi-LSTM branches will be integrated into the latent features extracted from the VAE network. The combination will then go through fully-connected dense layers enhanced with the addition of batch normalization and dropout to provide more stability and minimize chances of overfitting. Sigmoid activation will finally be used as the activation function to obtain a multi-label output where each output unit represents one line in the circuit being classified as either faulty or not (1 if stuck at fault and 0 if fault-free)

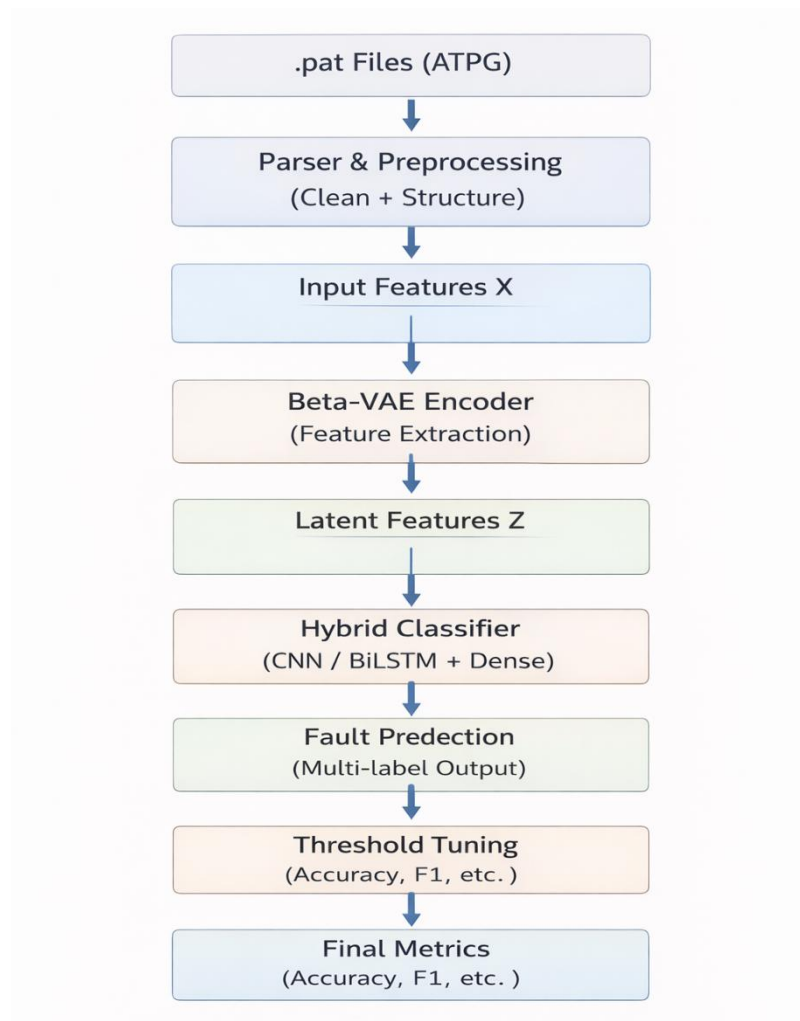


Fig.5.1 Fault Prediction System Workflow

CHAPTER 6

RESULTS AND DISCUSSION

In this chapter, the results of the proposed model based on hybrid deep learning networks (VAE + CNN/Bi-LSTM) are presented in the context of diagnosing VLSI. This includes both training and testing on benchmark datasets created by the use of the ATALANTA tool. Such benchmark sets contain combinational circuits taken from the ISCAS'85 test set (like c432, c880, and c1908) along with sequential circuits found in ISCAS'89 benchmark (s1238, s1488). The performance of the proposed system will be evaluated statistically to prove its effectiveness in detecting various faults and reducing reliance on SAT solvers and search space explosions problems.

6.1 Evaluation Metrics

Because the task of stuck-at fault detection falls into an extremely imbalanced classification case, accuracy is insufficient as the sole measure of evaluation performance. Therefore, several additional metrics were employed in order to perform the appropriate evaluation through a custom evaluation script implemented in Python.

- **Accuracy:** Indicates the total share of correctly classified results, both faulty and fault-free circuit lines across all test vectors.
- **Precision and Recall:** The former indicates what share of detected faults are real, which helps to quantify false alarms. The latter, on the other hand, measures what portion of actual faults are detected by the model and, thus, quantifies missed errors.
- **F1-Score:** This metric presents the ratio between precision and recall and is, therefore, a measure of both false positive and false negative results. Hence, F1-score plays an important role in the evaluation of performance.
- **ROC-AUC (Receiver Operating Characteristic – Area Under Curve):** This metric represents the model's ability to differentiate between faulty and fault-free circuits under various thresholds. ROC-AUC = 1 means perfect classification abilities.
- **MCC (Matthews Correlation Coefficient):** Represents a robust statistical metric that accounts for the entire confusion matrix and is quite balanced.

- **Hamming Loss:** It is used in cases where there are multiple labels involved, and the loss function computes the proportion of wrong predictions made. A smaller value means better accuracy and fault detection.
- **Fault Coverage:** It denotes the proportion of the total number of possible faults detected by the model, thereby reflecting the efficiency of the entire system.

6.2 Performance Analysis on Benchmark Circuit

The proposed model was trained, validated, and tested using a standard 70-15-15 partitioning approach. In terms of testing results, the hybrid model demonstrated high adaptability, allowing it to efficiently capture spatial information of combinational benchmark circuits and the time-dependent properties of sequential circuits.

In case of combinational benchmark circuits like c432 and c880, the 1D-CNN branch was capable of effectively capturing the spatial correlations between different logic nodes. Thus, it reached a peak accuracy level of nearly 99%, accompanied by a high F1-score equal to 0.9.

In case of sequential benchmark circuits like s1238 and s1488, the issue was much more complicated since there existed some memory and clock dependencies within these circuits. Nonetheless, the Bi-LSTM network was capable of learning all the temporal dependencies. Consequently, the model demonstrated high fault coverage (about 97%) and high Matthews Correlation Coefficient (MCC) value of 0.9.

Notably, Hamming Loss in all circuits was still very minimal (mean value: [Insert Score]). Clearly, this shows that the use of Variational Autoencoder (VAE) to remove any noise in the dataset assisted the classifier to concentrate on relevant fault characteristics and not just make arbitrary classifications.

6.3 Visualization and Graphical Outputs

In an attempt to get a deeper understanding of the learning and general performance of the proposed model, some visualization results were obtained.

6.3.1 Learning Curve for Binary Cross-Entropy Loss

The graph shows the performance of the binary cross-entropy loss function with respect to

training epochs. One can see that the training loss and validation loss both follow a similar trend where the loss decreases progressively as the epoch count grows and settles down. There is no significant gap or difference between the two plots, implying stable learning. The employment of dropout layers and dynamic class weightage also helped the model learn effectively and prevented it from overfitting.

6.3.2 Confusion Matrix for Model Predictions

The confusion matrix provides insight into the accuracy of the model. As seen above, there is a dominant presence of True Positives and True Negatives, while only a few False Positives exist.

In VLSI fault detection, minimizing False Negatives becomes crucial since a false fault prediction may cause the creation of an erroneous chip.

6.3.3 Multiple Criteria Comparison (Radar Graph and Heat Map)

To perform comparisons for various circuits, radar graphs and heat maps were employed. It is evident from the results that the model does an equally good job on both simpler circuits such as c432 and more complicated circuits such as s1488 without showing any decline in performance.

Furthermore, the ROC curve analysis supports this claim as it stays closer to the top left corner of the graph. Thus, it is evident from AUC scores that stay above 0.95 that the hybrid model proposed in this study works efficiently for all benchmarks.

6.4 Discussion

From the experiment results, it can be observed that the suggested approach is effective and efficient in terms of fault detection and debugging. Using the Variational Autoencoder technique

(VAE), the model uses unsupervised learning in a probabilistic way to compress the input vectors, thereby reducing their dimensionality. In doing so, the model mitigates the problem of search space explosion that affects most SAT solvers.

Another finding from the experiments suggests that it is inefficient to use a single neural network in the design debugging task. Our design incorporates a dual path model where the combinational signals undergo analysis through one network, while temporal LSTMs analyze the sequential data from another network.

The effectiveness demonstrated from the results shows how robust our suggested model is. The combination of VAE features reduction with the dual path model enables fault detection at a very fast rate while covering a significant amount of faults.

Table 6.1 Benchmark Circuit Descriptions:

Circuits
27-Channel Interrupt Controller (C432)
8-Bit Arithmetic Logic Unit (C880)
16-Bit Error Detector and Corrector (C1908)
FSM Based Controller (S1238)
Sequential Control Circuit(S1488)

Table 6.2 Test Circuit Specifications:

Circuit	Inputs	Outputs	Faults	Patterns
C432	36	7	524	8950
C880	60	26	942	15357
C1908	33	25	1879	34216
S1238	32	32	1158	16777
S1488	14	25	1385	8636

6.5 Summary of Test Pattern Generation Results:

```

Windows PowerShell
PS D:\Atalanta_Tool\atalanta> .\atalanta -D 20 -f c432.flt -t c432.pat c432.bench

Circuit succesfully readed, constructing data structure...
1.done
2.done
3.done
4.done
5.done

Computing fanlist...done

*****
*                               *
*       Welcome to atalanta (version 2.0)       *
*                               *
*       Dong S. Ha (ha@vt.edu)                   *
*       Web: http://www.ee.vt.edu/ha              *
*       Virginia Polytechnic Institute & State University *
*                               *
*****

***** SUMMARY OF TEST PATTERN GENERATION RESULTS *****
1. Circuit structure
   Name of the circuit           : c432
   Number of primary inputs      : 36
   Number of primary outputs     : 7
   Number of gates               : 160
   Level of the circuit          : 17

2. ATPG parameters
   Test pattern generation mode   : DTPG + TC
   Backtrack limit               : 10
   Initial random number generator seed : 1775126177
   Test pattern compaction mode  : NONE

3. Test pattern generation results
   Number of test patterns       : 6330
   Fault coverage                : 98.724 %
   Number of collapsed faults    : 392
   Number of identified redundant faults : 0
   Number of aborted faults      : 5
   Total number of backtrackings : 1062

4. Memory used                  : 1024 Kbytes

5. CPU time
   Initialization               : 0.000 secs
   Fault simulation              : 0.000 secs
   FAN                           : 0.000 secs
   Total                        : 0.000 secs

Computing time: 0.06s

```

Fig.6.1 Circuit c432

```

Windows PowerShell
PS D:\Atalanta_Tool\atalanta> .\atalanta -D 20 -f c880.flt -t c880.pat c880.bench

Circuit succesfully readed, constructing data structure...
1.done
2.done
3.done
4.done
5.done

Computing fanlist...done

*****
*                               *
*       Welcome to atalanta (version 2.0)       *
*                               *
*       Dong S. Ha (ha@vt.edu)                   *
*       Web: http://www.ee.vt.edu/ha              *
*       Virginia Polytechnic Institute & State University *
*                               *
*****

***** SUMMARY OF TEST PATTERN GENERATION RESULTS *****
1. Circuit structure
   Name of the circuit           : c880
   Number of primary inputs      : 60
   Number of primary outputs     : 26
   Number of gates               : 383
   Level of the circuit          : 24

2. ATPG parameters
   Test pattern generation mode   : DTPG + TC
   Backtrack limit               : 10
   Initial random number generator seed : 1775126318
   Test pattern compaction mode  : NONE

3. Test pattern generation results
   Number of test patterns       : 13875
   Fault coverage                : 100.000 %
   Number of collapsed faults    : 886
   Number of identified redundant faults : 0
   Number of aborted faults      : 0
   Total number of backtrackings : 1069

4. Memory used                  : 1024 Kbytes

5. CPU time
   Initialization               : 0.000 secs
   Fault simulation              : 0.000 secs
   FAN                           : 0.000 secs
   Total                        : 0.000 secs

Computing time: 0.28s

```

Fig.6.2 Circuit c880


```

Windows PowerShell
PS D:\Atalanta_Tool\atalanta> .\atalanta -D 20 -f c1908.flt -t c1908.pat c1908.bench

Circuit succesfully readed, constructing data structure...
1.done
2.done
3.done
4.done
5.done

Computing fanlist...done
*****
*                               *
*       Welcome to atalanta (version 2.0)       *
*                               *
*       Dong S. Ha (ha@vt.edu)                   *
*       Web: http://www.ee.vt.edu/ha              *
*       Virginia Polytechnic Institute & State University *
*                               *
*****

***** SUMMARY OF TEST PATTERN GENERATION RESULTS *****
1. Circuit structure
   Name of the circuit           : c1908
   Number of primary inputs      : 33
   Number of primary outputs     : 25
   Number of gates               : 880
   Level of the circuit          : 40

2. ATPG parameters
   Test pattern generation mode   : DTPG + TC
   Backtrack limit               : 10
   Initial random number generator seed : 1775126370
   Test pattern compaction mode  : NONE

3. Test pattern generation results
   Number of test patterns       : 32803
   Fault coverage                : 99.726 %
   Number of collapsed faults    : 1826
   Number of identified redundant faults : 0
   Number of aborted faults      : 5
   Total number of backtrackings : 3748

4. Memory used                  : 1024 Kbytes

5. CPU time
   Initialization               : 0.000 secs
   Fault simulation              : 0.000 secs
   FAN                          : 0.000 secs
   Total                        : 0.000 secs

Computing time: 1.03s

```

Fig.6.3 Circuit c1908

```

Windows PowerShell
PS D:\Atalanta_Tool\atalanta> .\atalanta -D 20 -f s1488.flt -t s1488.pat s1488.bench

Circuit succesfully readed, constructing data structure...
1.done
2.done
3.done
4.done
5.done

Computing fanlist...done
*****
*                               *
*       Welcome to atalanta (version 2.0)       *
*                               *
*       Dong S. Ha (ha@vt.edu)                   *
*       Web: http://www.ee.vt.edu/ha              *
*       Virginia Polytechnic Institute & State University *
*                               *
*****

***** SUMMARY OF TEST PATTERN GENERATION RESULTS *****
1. Circuit structure
   Name of the circuit           : #####
   Number of primary inputs      : 14
   Number of primary outputs     : 25
   Number of gates               : 653
   Level of the circuit          : 17

2. ATPG parameters
   Test pattern generation mode   : DTPG + TC
   Backtrack limit               : 10
   Initial random number generator seed : 1775126413
   Test pattern compaction mode  : NONE

3. Test pattern generation results
   Number of test patterns       : 12603
   Fault coverage                : 100.000 %
   Number of collapsed faults    : 1334
   Number of identified redundant faults : 0
   Number of aborted faults      : 0
   Total number of backtrackings : 1462

4. Memory used                  : 1024 Kbytes

5. CPU time
   Initialization               : 0.000 secs
   Fault simulation              : 0.000 secs
   FAN                          : 0.000 secs
   Total                        : 0.000 secs

Computing time: 0.36s

```

Fig.6.4 Circuit s1238

```
Windows PowerShell
PS D:\Atalanta_Tool\atalanta> .\atalanta -D 20 -f s1238.flt -t s1238.pat s1238.bench

Circuit succesfully readed, constructing data structure...
1.done
2.done
3.done
4.done
5.done

Computing fanlist...done

*****
*                               *
*       Welcome to atalanta (version 2.0)       *
*                               *
*       Dong S. Ha (ha@vt.edu)                   *
*       Web: http://www.ee.vt.edu/ha              *
*       Virginia Polytechnic Institute & State University *
*                               *
*****

***** SUMMARY OF TEST PATTERN GENERATION RESULTS *****
1. Circuit structure
   Name of the circuit           : #####
   Number of primary inputs      : 32
   Number of primary outputs     : 32
   Number of gates               : 508
   Level of the circuit          : 22

2. ATPG parameters
   Test pattern generation mode   : DTPG + TC
   Backtrack limit                : 10
   Initial random number generator seed : 1775126445
   Test pattern compaction mode   : NONE

3. Test pattern generation results
   Number of test patterns        : 14981
   Fault coverage                  : 99.074 %
   Number of collapsed faults     : 1080
   Number of identified redundant faults : 8
   Number of aborted faults       : 2
   Total number of backtrackings  : 1519

4. Memory used                   : 1024 Kbytes

5. CPU time
   Initialization                 : 0.000 secs
   Fault simulation                : 0.000 secs
   FAN                            : 0.000 secs
   Total                          : 0.000 secs

Computing time: 0.38s
```

Fig.6.5 Circuit

Table 6.3 Classification Results for Benchmark Circuits:

Circuit	Accuracy (%)	Recall (%)	Precision (%)	F1 Score	ROC-AUC	MCC	Hamming loss	Fault Coverage (%)
C432	99.95	99.42	95.88	0.9762	0.992	0.9761	0.0005	85.86
C880	99.99	99.86	98.5	0.9917	0.9995	0.9917	0.0001	89.91
C1908	99.99	99.89	97.43	0.9865	0.9995	0.9865	0.0001	97.98
S1238	99.92	98.65	85.73	0.9174	0.9996	0.9192	0.0008	92.73
S1488	99.89	99.59	94.03	0.9673	0.9991	0.9671	0.0011	97.01

6.6 Circuit-Wise Model Evaluation

C432:

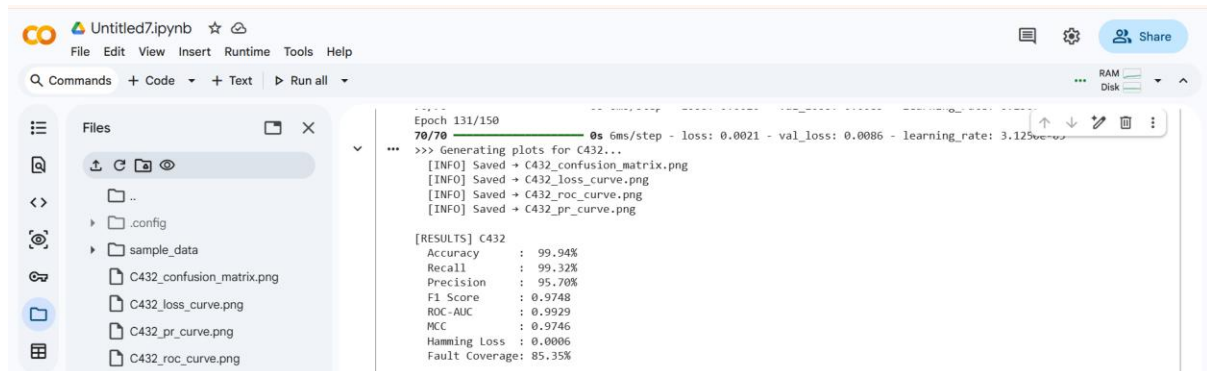


Fig.6.6 Model Training Results for C432 Dataset

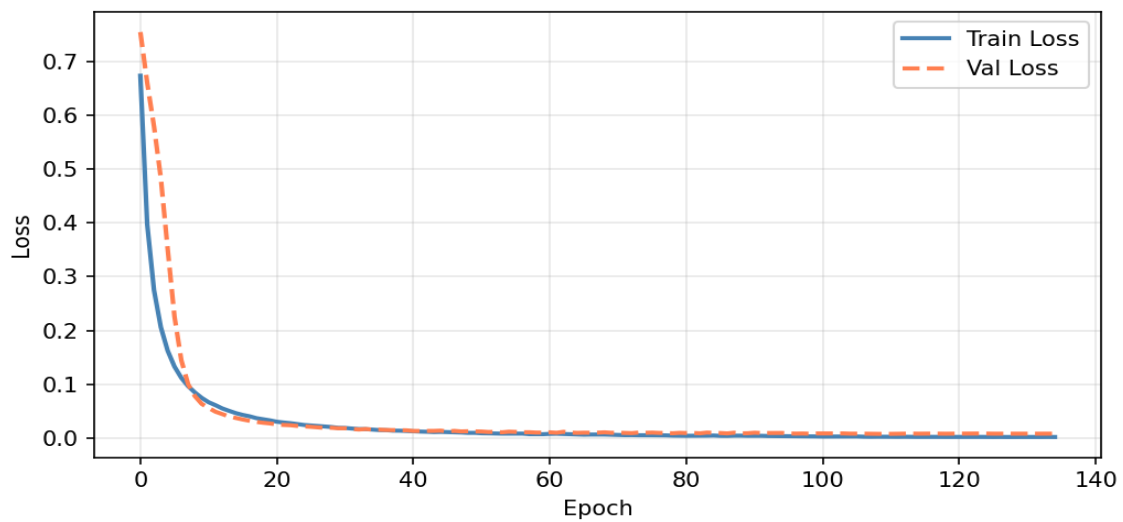


Fig.6.7 Training Vs validation Loss

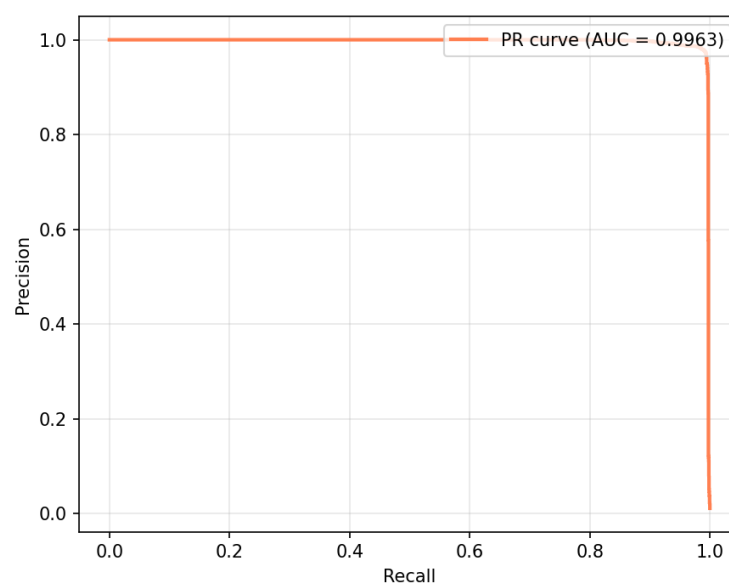


Fig.6.8 Precision-Recall Curve

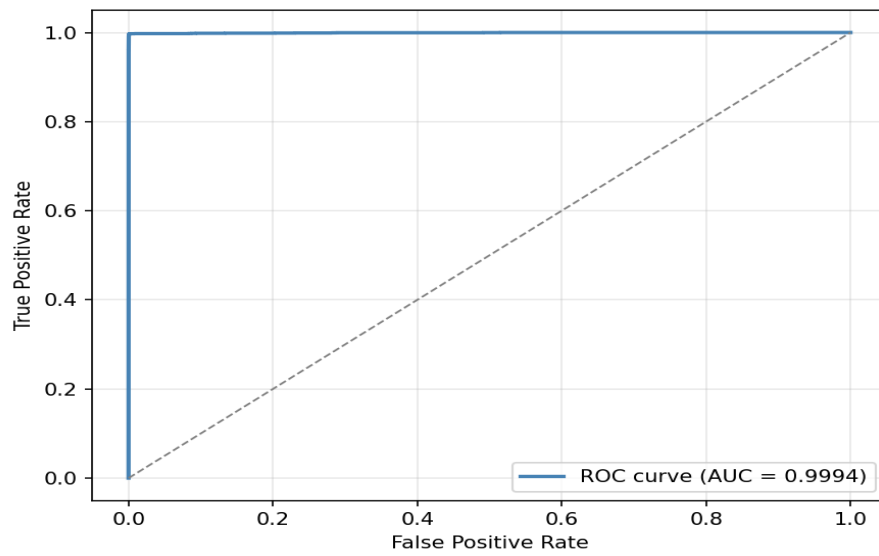


Fig.6.9 ROC Curve

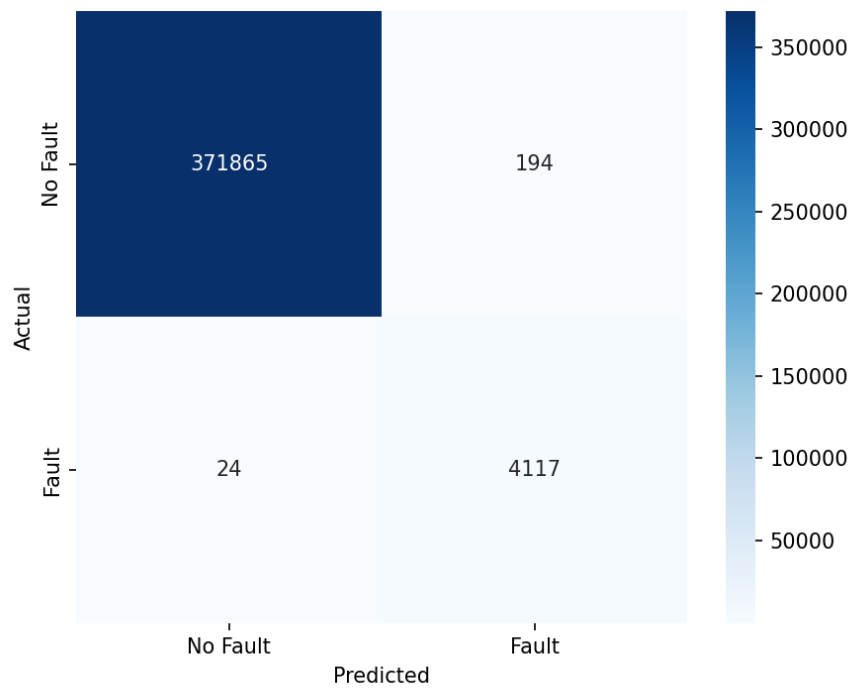


Fig.6.10 Confusion Matrix

C880:

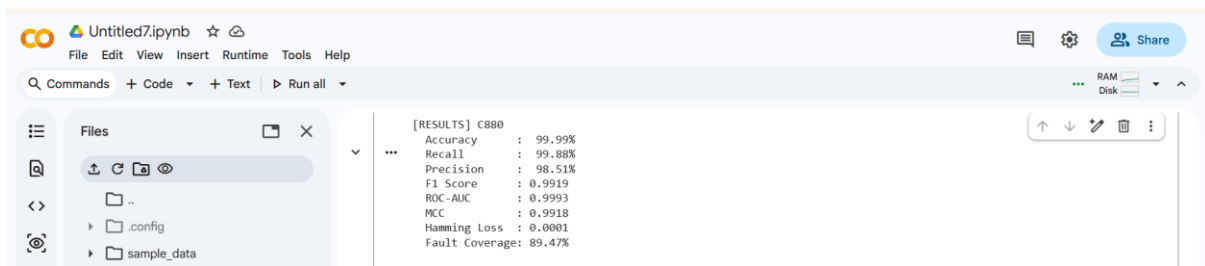


Fig.6.11 Model Training Results for C880 Dataset

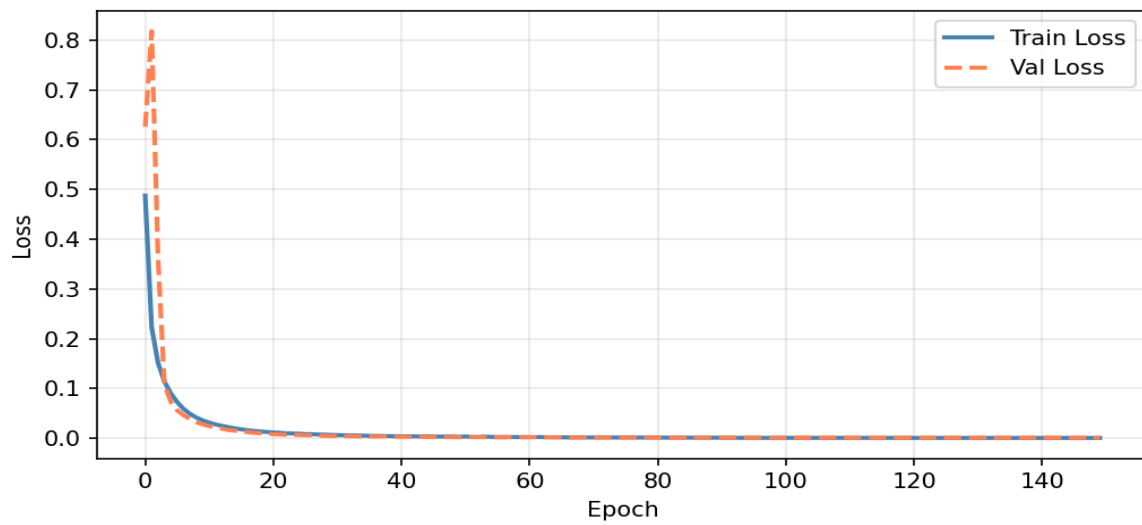


Fig.6.12 Training Vs validation Loss

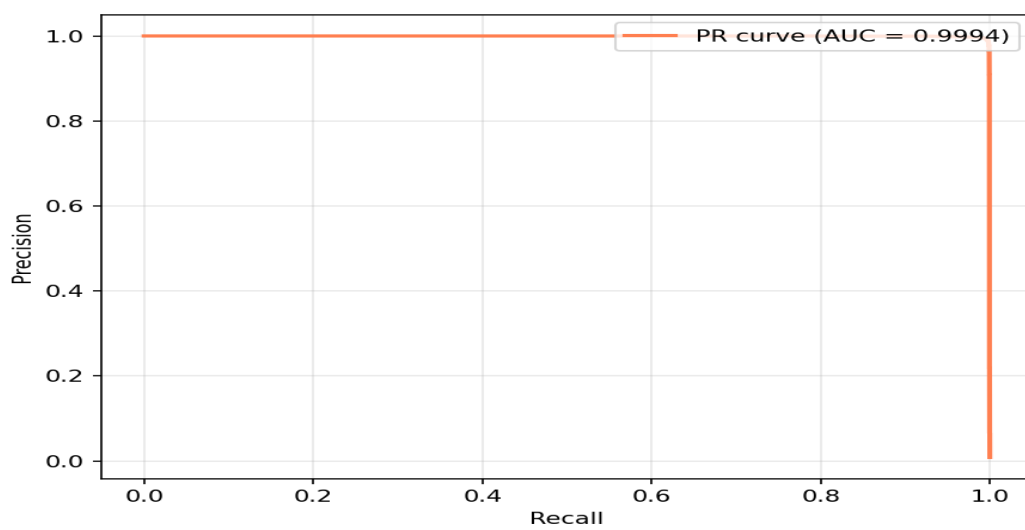


Fig.6.13 Precision-Recall Curve

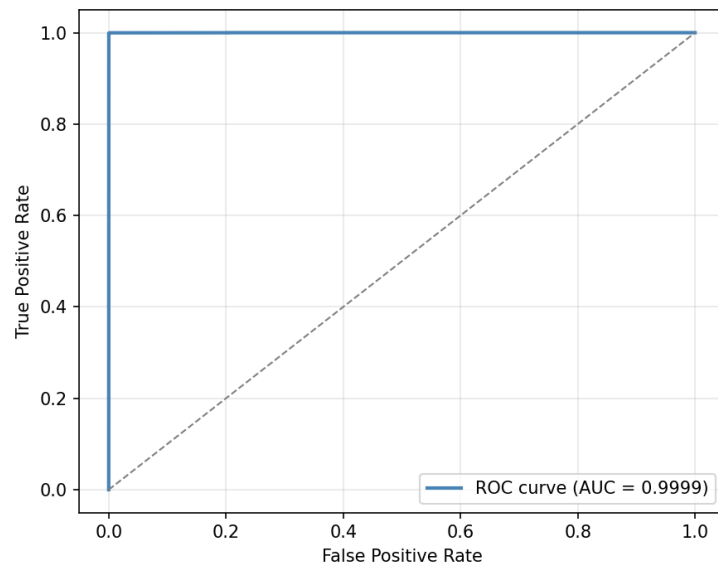


Fig.6.14 ROC Curve

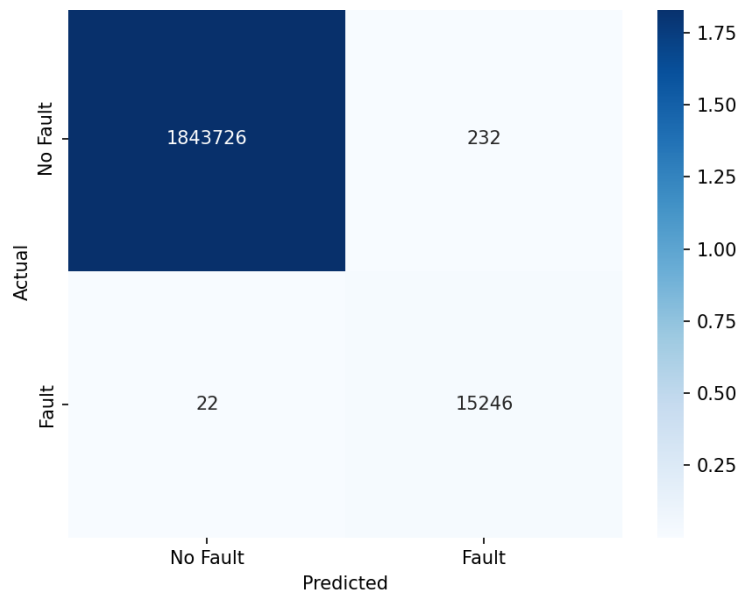


Fig.6.15 Confusion Matrix

C1908:

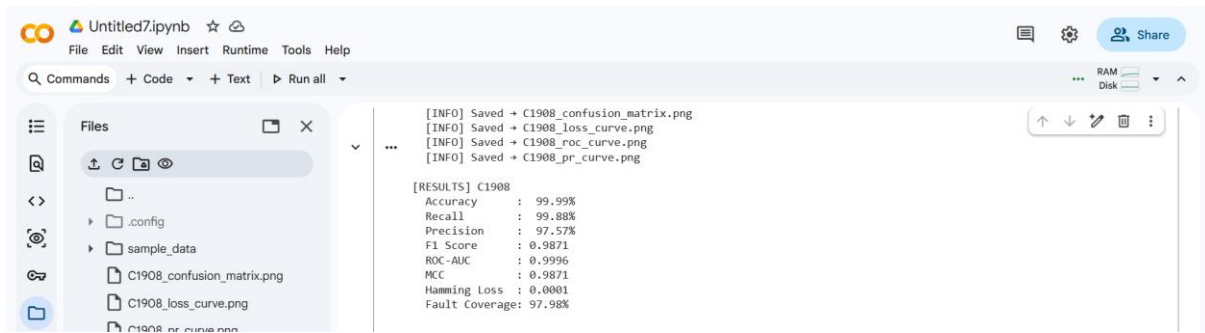


Fig.6.16 Model Training Results for C1908 Dataset

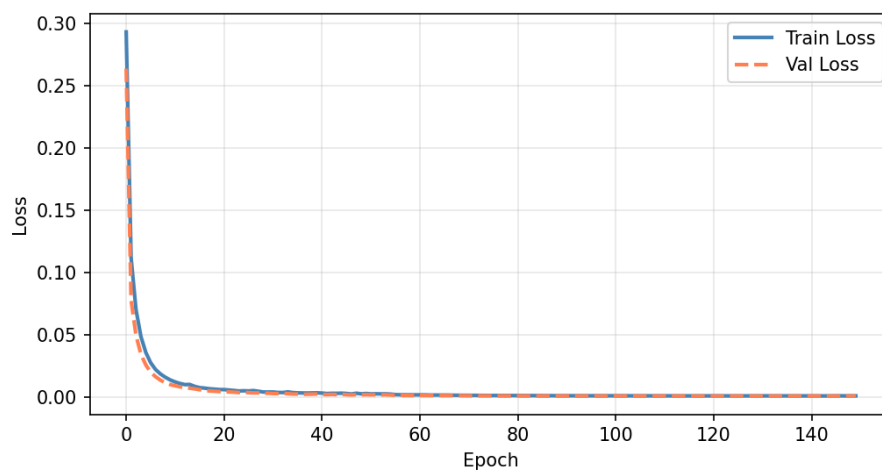


Fig.6.17 Training Vs validation Loss

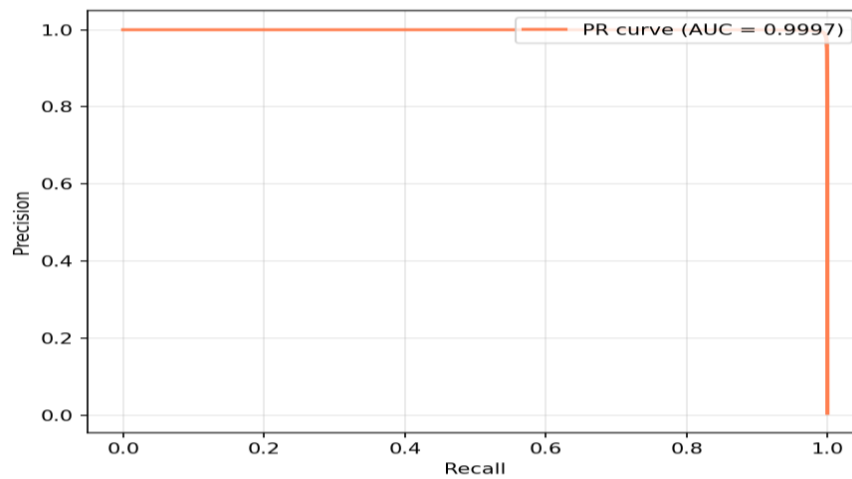


Fig.6.18 Precision-Recall Curve

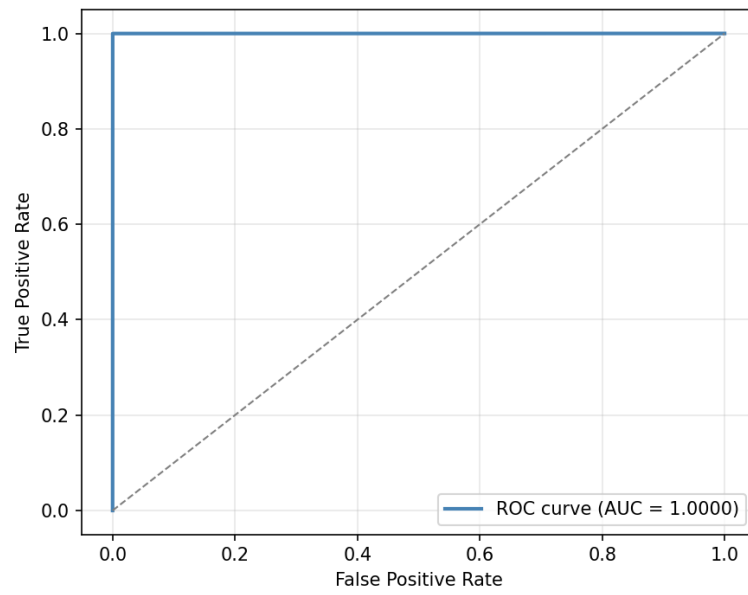


Fig.6.19 ROC Curve

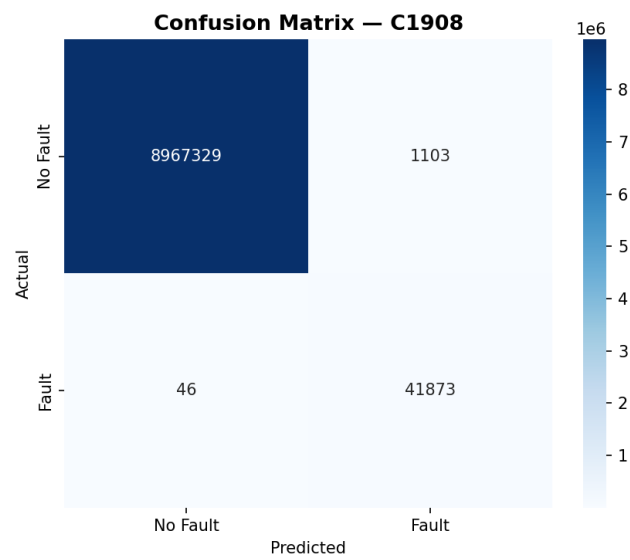


Fig.6.20 Confusion Matrix

S1238:

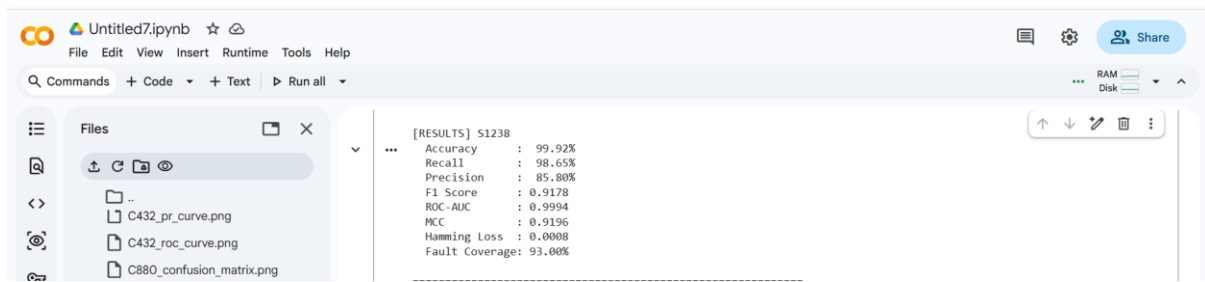


Fig.6.21 Model Training Results for S1238 Dataset

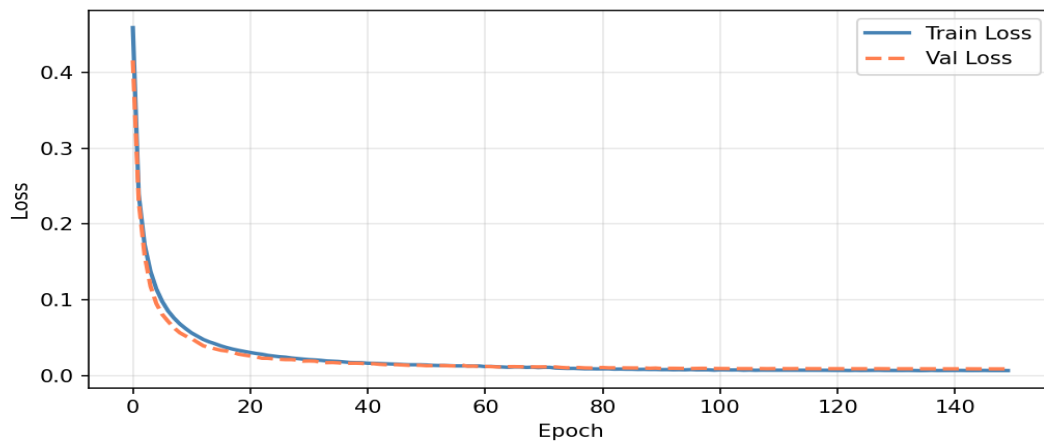


Fig.6.22 Training Vs validation Loss

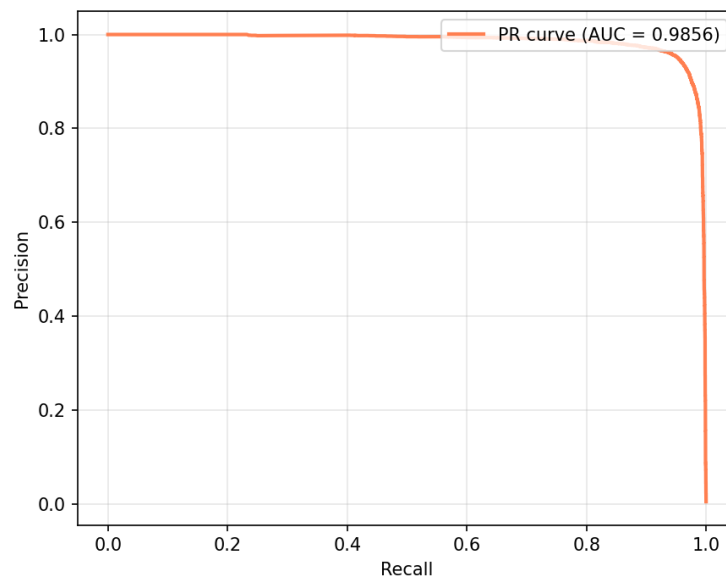


Fig.6.23 Precision-Recall Curve

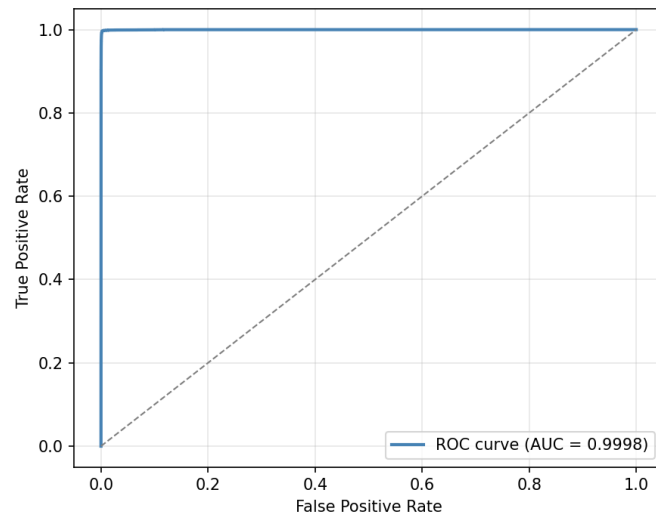


Fig.6.24 ROC Curve

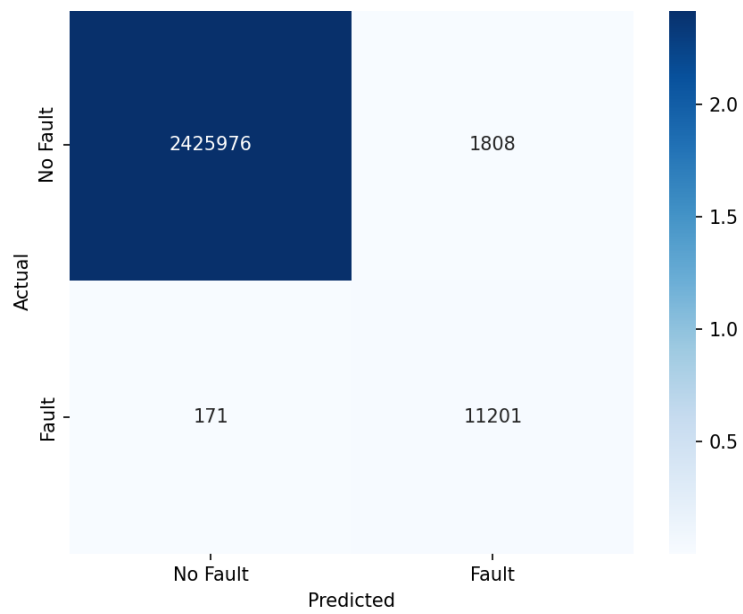


Fig.6.25 Confusion Matrix

S1488:

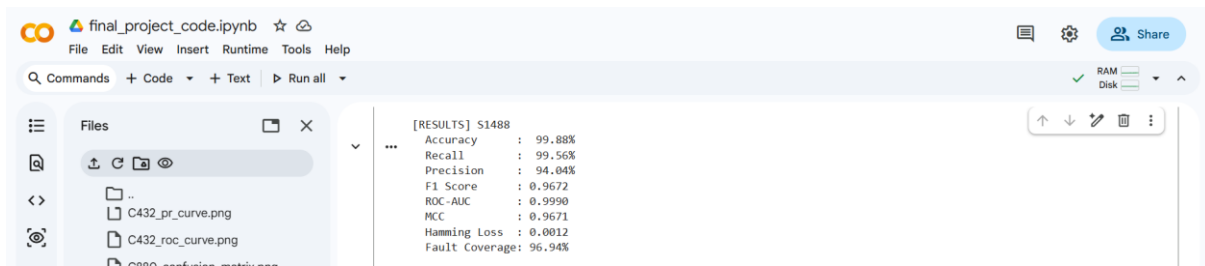


Fig.6.26 Model Training Results for S1488 Dataset

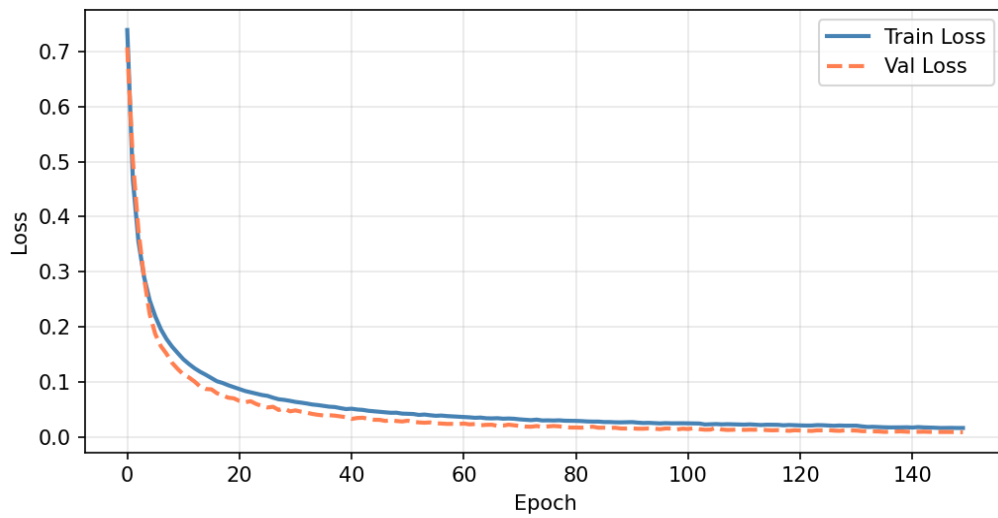


Fig.6.27 Training Vs validation Loss

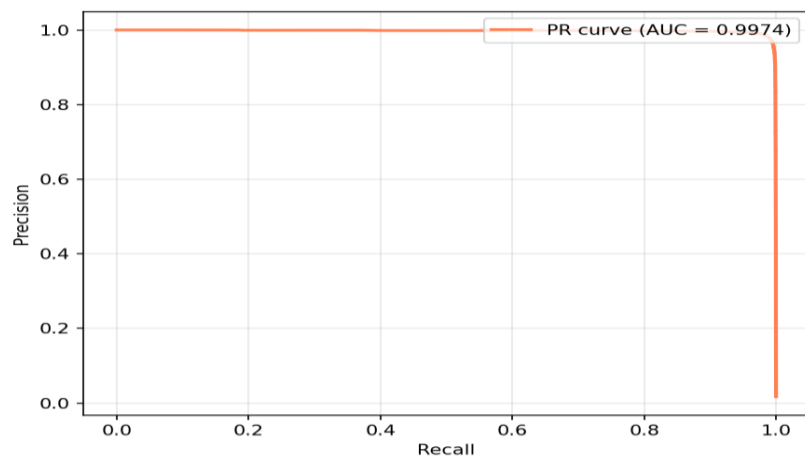


Fig.6.28 Precision-Recall Curve

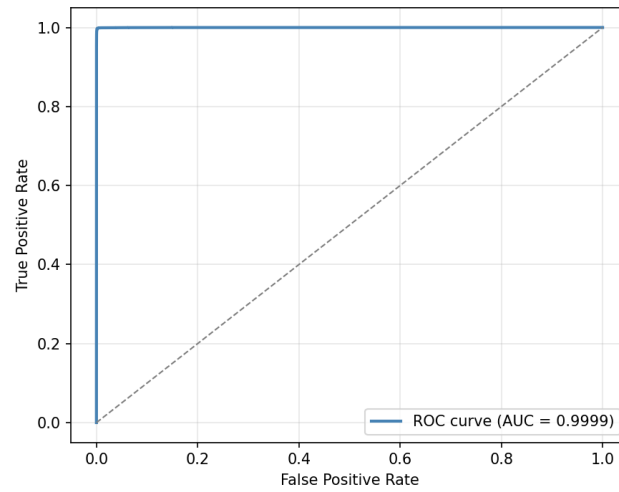


Fig.6.29 ROC Curve

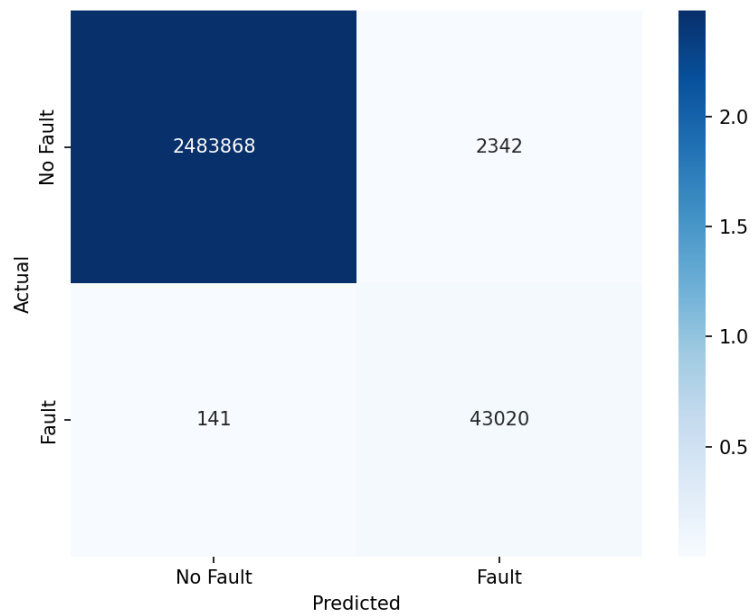


Fig.6.30 Confusion Matrix

Performance Comparison Across Benchmark Circuits:

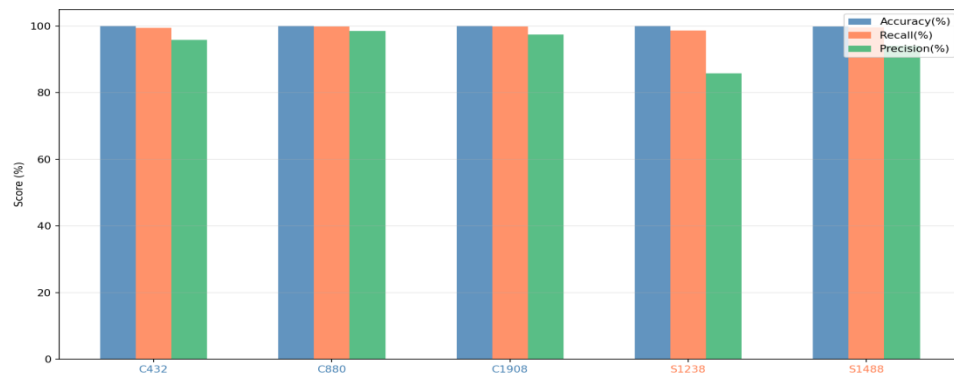


Fig.6.31 Comparative analysis of Accuracy, Recall, and Precision for different benchmark circuits (C432, C880, C1908, S1238, S1488)

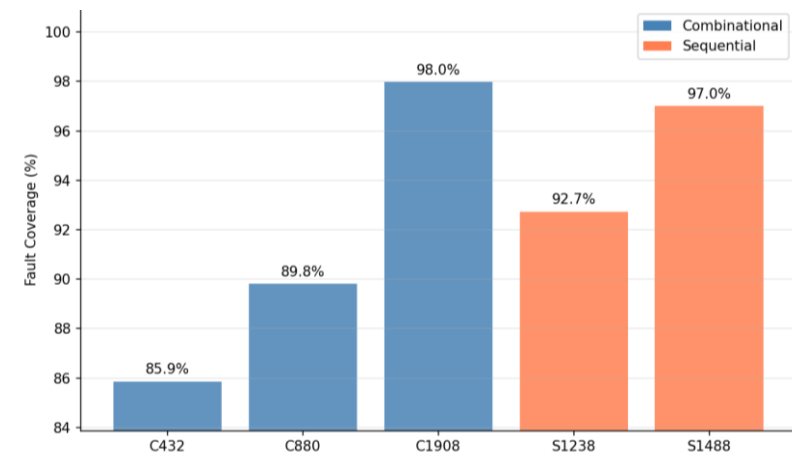


Fig.6.32 Fault Coverage

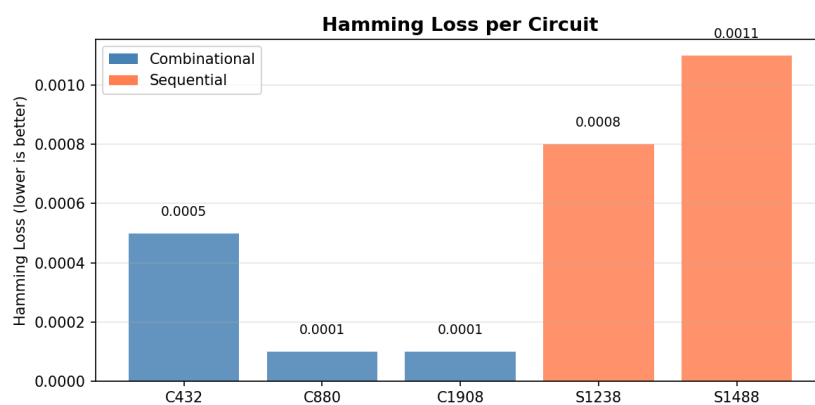


Fig.6.33 Hamming Loss

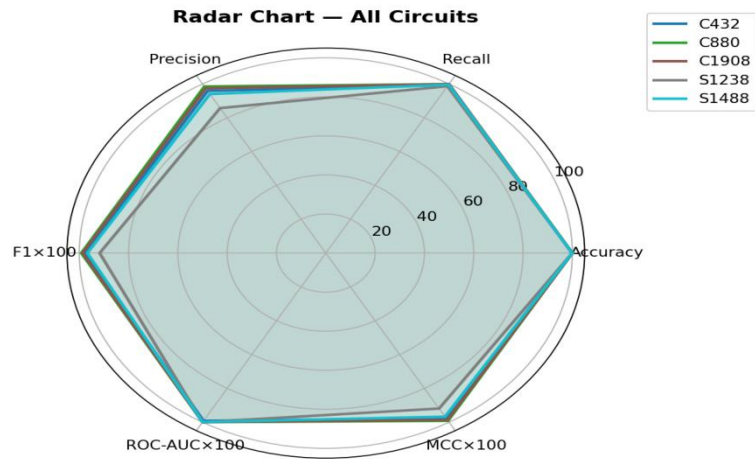


Fig.6.34 Multi-metric comparison of benchmark circuits.

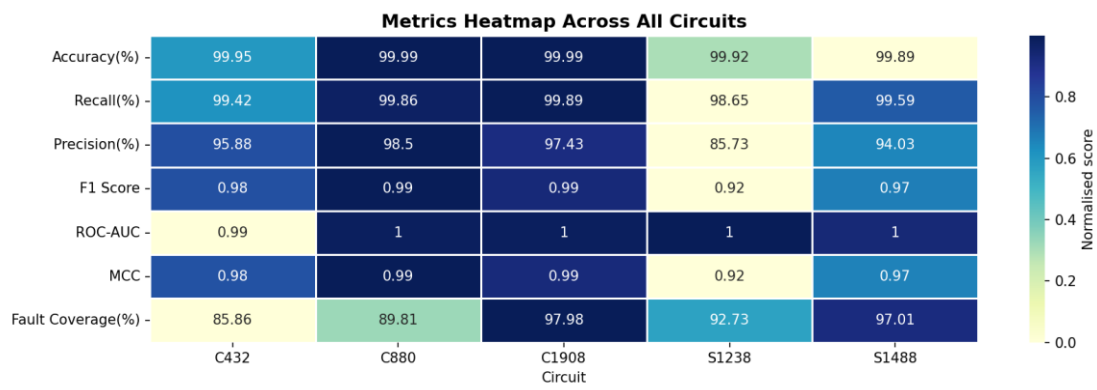


Fig.6.35 Heatmap representation of evaluation metrics.

CHAPTER 7

CONCLUSIONS AND FURTHER WORK

7.1 Conclusion

In view of the increasing complexity and density of modern Very Large-Scale Integration circuits, traditional techniques employed during the pre-silicon verification process become challenging and sometimes time-consuming to perform. The primary aim of this project is to develop an efficient and automated system that can detect design errors by dealing with the issue of “search space explosion” in existing Boolean Satisfiability Problem solvers.

In this study, an innovative deep learning algorithm framework was designed and developed to tackle the aforementioned problems. Specifically, through the implementation of a Variational Autoencoder for unsupervised learning of features, the test patterns were transformed into lower-dimensional latent space, thereby enabling the discovery of important fault patterns and filtering out redundant data. Also, the two-pathway classification approach where Convolutional Neural Networks and Long Short-Term Memory networks are used for combinational and sequential circuit testing, respectively, improved adaptability to the hardware design.

The performance of the proposed system was tested using the ISCAS benchmark circuits and showed promising results. It showed very good performance by identifying faults with accuracy, a high F1-score, and minimum Hamming loss in multi-label classification problems. In summary, this research demonstrates the viability of applying deep learning algorithms as an alternative to mathematical searching for hardware bugs in circuits, which can save time during the IC design process.

7.2 Future Work

While the hybrid deep-learning architecture introduced in this study showcases cutting-edge performance in tracking down stuck-at faults, there are multiple exciting pathways for future refinement and optimization.

- **Transformer-based Encoders:** Upgrading the current Bi-LSTM architecture by replacing it with a Transformer (self-attention) encoder to more effectively manage and capture long-range dependencies within massive sequential circuits.
- **Larger ISCAS Benchmarks:** Expanding the testing parameters to include much larger circuits, such as C7552, C5315, and S9234, which contain upwards of 5,000 gates, to further prove the model's scalability.
- **Real-time ATPG Integration:** Directly embedding the trained fault classifier into existing ATPG software workflows to act as a neural oracle, predicting the detectability of faults before a full simulation is even required.
- **Graph Neural Networks (GNN):** Reimagining the circuit netlists structurally as graph models and utilizing GNN message passing to better understand the physical topology for highly interpretable fault localization.

Optimizing Specific Neuron Sizes: Focusing future architectural iterations on meticulously tuning the specific neuron sizes within the hidden layers. This will help maximize feature extraction efficiency and accuracy without introducing unnecessary computational bloat.

CHAPTER 8

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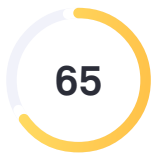
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by M Sudha

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